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Associations between Graduated Driver Licensing restrictions and delay in driving licensure among U.S. high school students

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A R T I C L E I N F O

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A B S T R A C T

Introduction: Some of the most vulnerable groups of teens choose to delay driving licensure (DDL). We assessed longitudinal associations between state-level Graduated Driver Licensing (GDL) restrictions and DDL among U.S. high school students.
Methods: Data from seven waves of the NEXT Generation Health Study (starting 10th-grade (2009–2010)), were analyzed in 2020 using Poisson regression. The outcome was DDL (delay vs. no-delay). Independent variables were driving restrictions (at learner and intermediate phases of licensure), sex, race/ethnicity, family affluence, parent education, family structure, and urbanicity.
Results: Of 2525 eligible for licensure, 887 (38.9%), 1078 (30.4%), 560 (30.7%) reported DDL 1–2 years, >2 years, no DDL, respectively. Interactions between GDL restrictions during the learner permit period and covariates were found. In states requiring ≥30 h of supervised practice driving, Latinos (Adjusted relative risk ratio [aRRR] = 1.55, $p < .001$) and Blacks (aRRR = 1.38, $p < .01$) were more likely to DDL than Whites. In states where permit holding periods were <6 months, participants with low (aRRR = 1.61, $p < .001$) and moderate (aRRR = 1.45, $p < .001$) vs. high affluence were more likely to DDL. Participants in single-parent households vs. both-biological parent households were also more likely to DDL (aRRR = 1.37, $p < .05$). In states where permit holding periods were ≥6 months, participants with low (aRRR = 1.33, $p < .05$) vs. high affluence were more likely to DDL. In states that allowed ≥3 passengers or no passenger restriction, participants living in non-urban vs. urban (aRRR = 1.52, $p < .05$) areas were more likely to DDL, and in states that allowed only 1 or no passenger, participants living in non-urban vs. urban areas (aRRR = 0.67, $p < .001$) were less likely to DDL.
Conclusions: Our findings heighten concerns about increased crash risk among older teens who age out of state GDL policies thereby circumventing driver safety related restrictions. Significant disparities in DDL exist among more vulnerable teens in states with stricter GDL driving restrictions.

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1. Introduction

Graduated Driver Licensing (GDL) programs were put in place to introduce new young drivers gradually and safely into the driving population, gaining needed driving experience before obtaining full independent driving privileges (Foss et al., 2001; Governors Highway Safety Association, n.d.). It has been well documented that comprehensive GDL programs are associated with reduced fatal crash involvements by 20–40% among teen drivers in the U.S. (Chen et al., 2006; Shope, 2007; Williams, 2007; Zhu et al., 2009; Zhu et al., 2013) as well as in other countries such as Australia, New Zealand, Northern Ireland, and Canada (Begg and Stephenson, 2003; Christie et al., 2017; Scott-Parker, 2016). In the U.S., every state has a GDL program with three stages, a learner's permit, intermediate license, and unrestricted license (Centers for Disease Control and Prevention (CDC), 2016; Masten et al., 2013). In 1996, the adoption of GDL policies in all states began with the incorporation of additional novice driver requirements and restrictions (e.g., mandated supervised practice driving hours, limiting night time driving and number of passengers) during the learning period (Williams et al., 2016).

Across the U.S., the minimum age to obtain a driver permit can range from 14 to 16 years old, a restricted license from 14 to 17 years old, and a full license typically from 16.5 to 18 years old (Williams et al., 2016; Williams, Tefft and Grabowski, 2012b). Previous research has shown that GDL laws requiring older minimum licensing age were associated with reduced crash rates among young drivers (McCartt et al., 2010). This has been attributed to an age effect in several studies (Mayhew et al., 2003; McCartt et al., 2009). Alternatively, extended permit holding periods try to increase safety by providing youth with greater and longer opportunity for driving exposure (i.e., driving practice). As of 2010, 47 states require new drivers to hold a permit ranging from 6 to 12 months before obtaining a license (Williams, Tefft and Grabowski, 2012a). A five-year survival analysis study conducted in North Carolina found a permit holding period of 9–12 months to reduce crash rates among teen drivers (Masten and Foss, 2010). Furthermore, a New Zealand study found young drivers who gained a license after 18 months had significantly less crash involvement compared to drivers who gained their license after a holding period of 12–18 months (Lewis-Evans, 2010). Even so, a study analyzing collision claim frequencies in young drivers found that mandatory holding periods for a learner's permit did not reduce crash risk once they were licensed when age was not taken into consideration (Trempel, 2009). This suggests a reduction in crash frequency could be due to drivers simply being older or greater pre-independent licensure driving experience. Generally, most states require 40–50 h of supervised practice driving for the young driver by a parent or guardian as a part of their GDL program (Foss, Masten, Goodwin, O'brien, & TransAnalytics, 2012). However, this is a source of controversy given that adult supervisors may choose to "sign off" on the total number of required practice-driving hours as completed even if they were not. A naturalistic driving study in Virginia found 54% of young drivers did not truly complete all supervised hours before obtaining their driver's license (Ehsani et al., 2017). It is possible that low adherence to required supervised practice driving hours contributed to research findings of no independent association between fatal crash rates among young drivers and increasing required practice hours (Masten et al., 2015; McCartt et al., 2010).

In the intermediate license stage, common GDL restrictions include passenger and nighttime driving restrictions. These restrictions are in place to allow young drivers to continue to practice driving and continue to gain experience mainly under lower crash risk conditions. Particularly for young drivers, there is a significant increase in more severe and fatal when passengers are present (Ouimet et al., 2015). Teenage passengers seem to pose the biggest risk to a vehicle crash when the driver is a teenager (Tefft et al., 2012). The negative effect is often attributed to increased distraction from teenage passenger behaviors such as talking loudly, and sudden physical movements (e.g., "rough housing"). Recent research has found passenger restrictions to decrease teenage crash involvement (Masten et al., 2015; McCartt and Teoh, 2015). Nighttime driving restrictions have also been found to reduce crashes and injuries among teenage drivers (McCartt et al., 2010; Williams et al., 2012b). Approximately one third of 16-to-17 y/o drivers involved in a crash in the U.S. between the hours of 9:00 p.m. and 5:59 a.m. (Shults, 2016). This high crash risk is often attributed to drowsy driving and alcohol use during the later night to early morning hours.

Recently, the rate of teens that delay driving licensure (DDL) has increased. Between 1996 and 2010 the percentage of U.S. high school seniors with a driver's license decreased from 83% to 73% (Shults and Williams, 2013). This trend is concerning as GDL programs have been established to decrease overall crash risk in novice drivers but most of these programs only extend to drivers at the age of 18 years and younger (Curry et al., 2018). As of 2017, only 7 states required some GDL restrictions for novice drivers ≥18 years old (Curry et al., 2017). A study conducted in New Jersey found that older novice drivers (i.e., ≥21years) experience fewer crashes compared to younger novice drivers (age 17–20 years) under GDL policies (Curry et al., 2017). However, another study found at-fault crashes to be positively associated with driver age at the time of licensure (Waller et al., 2001), so there are inconsistencies in the findings. While DDL (i.e., generally referring to drivers that age-out of state GDL policy) may lead to less driver crash risk due to the reduced driving exposure and restricting driving under high-risk conditions (Hartos et al., 2001), this reduction may be reversed later among youth that DDL and end up with insufficient driving exposure, experience, and instruction (Tefft et al., 2013) because of missing important GDL safety practices. Black and Latino youth engage in DDL at higher rates than their White counterparts (Shults and Williams, 2013; Tefft et al., 2014). Furthermore, youth with low income backgrounds are more likely to delay licensure compared to high income youth (Curry et al., 2015; Tefft et al., 2014; Thigpen and Handy, 2018).

Currently, there is conflicting evidence on how GDL programs may be associated with DDL. A retrospective case study (Thigpen and Handy, 2018) investigated the factors influencing the increasing DDL among young adults and found that multiple factors including travel attributes, driver attitudes, parental influences, and GDL policies, have considerable influence on license possession and timing. Other studies have found GDL to have a slight influence on DDL among youth (Tefft et al., 2014). On the other hand, Australian researchers have found that rates of DDL had been increasing prior to the establishment of their GDL programs, with no significant increases after GDL was established (Raimond and Milthorpe, 2010). Given the mixed findings, the purpose of this study was to assess longitudinal associations of state-level GDL restrictions with DDL among U.S. high school students.

2. Methods

2.1. Design, setting, and participants

2.1.1. Design

Data used in this project were from all seven waves (W1-7) of the longitudinal NEXT Generation Health Study (NEXT), a longitudinal study that followed a nationally representative cohort of U.S. 10^{th} grade students (mean age = 16.3 y/o (se = 0.03)) adolescents into emerging adulthood. Additional information about the NEXT study can be found elsewhere (Li et al., 2013; Vaca et al., 2020).

2.1.2. Setting

This study used primary sampling units (PSU, school districts) from the nine U.S. census divisions including 22 states: CA, CO, CT, FL, GA, ID, IL, KY, LA, MA, MI, MN, NV, NY, OH, OR, PA, SD, TN, TX, WI, and WV. Schools and classrooms were randomly sampled from the PSUs.

2.1.3. Participants

A total of 2785 students from 81 high schools participated in the NEXT study. W1 data collection began in the 2009–2010 school year and continued to survey participants yearly until W7 in 2016. From W1–W7, 91%, 88%, 86%, 78%, 79%, 84%, and 83% of the full sample (N = 2785) completed the survey during the springtime each year. Parental consent and adolescent assent were obtained. When participants turned 18 years of age, their consent was obtained. The protocol for the study was approved by the Institutional Review Board of the *Eunice Kennedy Shriver* National Institute of Child Health and Human Development. Analyses were completed in 2020.

2.2. Measures

2.2.1. Outcome variable

The dichotomous outcome variable was delayed driving licensure (DDL). At each annual assessment the NEXT survey repeated the following question, “Do you have a driver’s license?”, with the possible responses being, No license of any sort; Permit to take the classroom component driver education only; Permit allowing supervised practice driving with an instructor or licensed adult; License allowing independent, unsupervised driving (with or without temporary restrictions on late night driving, teen passengers, etc.). In this study, we only included participants who had a license allowing independent-unsupervised driving, and DDL is defined as the time in years elapsed between the year that a participant was legally eligible to be licensed and the age they were licensed. Specifically, DDL was calculated as year(s) elapsed between participant’s eligible year and licensed year, or eligible year and final assessment if not licensed and then dichotomized (No Delay vs. Delay).

2.3. Independent variables

2.3.1. GDL restrictions

We used state-level GDL law details per the Governors Highways Safety Association (Governors Highway Safety Association, n.d.) for the period when NEXT data were collected and GDL law strength was determined by a point-rating system where key safety-related licensing components were scored as described by McCartt et al. (2010). GDL components were coded such that zero points indicate no or low GDL restrictions and one or more points indicates higher restriction. Restrictions on the learner’s phase were coded as: minimum permit age (0 points = less than 16 years, 1 point = 16 years or older), permit holding period (0 points = less than 6 months, 1 point = 6 or more months), and permit required practice driving hours (0 points = less than 30 h, 1 point = 30 or more hours). Restrictions in the intermediate phase were coded as: limits on night driving points (0 points = no restriction, 1 point = after 10 p.m., 2 points = 10 p.m. or earlier) and limits on young (age of 18–25 or younger, depending on state) passenger(s) allowed (0 points = 3 or more passengers or no restriction, 1 point = 2 passengers, 2 points = zero or 1 passenger).

2.3.2. Covariates

Covariates assessed in this study included sex (male vs. female), race/ethnicity (Latino, Non-Latino Blacks, others vs. Non-Latino Whites), family affluence, parental education, and onset age of initial alcohol use.

2.3.3. Family affluence

Family socioeconomic status (SES) was estimated using the validated Family Affluence Scale (Currie et al., 2004) including number of vehicles (i.e., including car, van, and/or truck) owned by family (none = 0, one = 1, and ≥2 = 2), number of computers owned by family (none = 0, one = 1, two = 2, >2 = 2), whether the student had his/her own bedroom (no = 0, yes = 1), and the number of family vacations in the last 12 months (not at all = 0, once = 1, twice = 2, >twice = 2). A composite score was calculated for each young person based on his or her responses to these four items and students were then categorized as low (score = 0–3), moderate (score = 4, 5) and high affluence (score = 6, 7) (Spriggs et al., 2007).

2.3.4. Family structure

Students' family structure was categorized as: both biological parents; one biological parent, one stepparent; single parent, mother only; single parent, father only; and other.

2.3.5. Parental education

We categorized the higher education level of both parents as less than high school diploma; high school diploma or general equivalency diploma (GED); some college, technical school, or associate degree; and bachelor's or graduate degree.

2.3.6. Onset age of initial alcohol use

Onset age of initial alcohol use was measure by asking participants, "How old were you the first time you had a drink of an alcoholic beverage? (Please do not include any time when you only had a sip or two from a drink.)" at W1 to W7. The reported year at the earliest wave was used. Then, the participants were categorized into two group (≥14 vs. <14 y/o).

2.3.7. Urbanicity

Urbanicity was defined as participants' school location at W1 using a seven-level scale ranging from metropolitan city to rural region. Those attending schools in an urban region were categorized as urban, and remaining categories were classified as non-urban (i.e., suburban/rural).

2.3.8. Statistical analysis

Due to the fact that the DDL is a common (vs. rare) variable, Poisson regressions were conducted to assess the associations between the DDL and each GDL restriction (Cummings, 2009; Greenland, 2004). Valid confident intervals (CIs) were estimated with a robust estimator of variance (Wolter, 2007) and risk ratios (i.e., exponentiated coeffects) were estimated to indicate the risk (prevalence ratio) of DDL compared with risk (prevalence ratio) of no DDL (Cummings, 2009). Separate Poisson models were run for each demographic variable (race/ethnicity, sex, family affluence, family structure, parent education, and onset age of alcohol use) that included the interaction term of each demographic variable by each GDL restriction, controlling for other demographic variables. The association between DDL and demographic variables by GDL restrictions were reported (Table 3) if the interaction was statistically significant. Interaction contracts were used to examine the significant associations; for example, given states that required 30 or more hours of supervised practice driving, likelihoods of DDL in Latinos and in Non-Latino Blacks compared to the likelihood of DDL in Non-Latino Whites (as the reference) was calculated respectively and denoted by adjusted relative risk ratio (aRRR). Missing data were deleted listwise. We performed the Poisson regressions with Stata version 16 (StataCorp LLC, TX) and other analyses with SAS version 9.4 (SAS Institute, Cary, NC) and accounted for features of the complex survey design (i.e., stratification, clustering, and sampling weights) in all

Table 1
Sample characteristics.

	Overall		DDL		No DDL	
	N	Weighted %	N	Weighted %	N	Weighted %
Race/ethnicity						
Latino	835	19.3	684	89.9	42	10.1
Non-Latino Black	687	20.2	567	83.7	57	16.3
Non-Latino White	1106	55.7	608	57.7	438	42.3
Other	142	4.8	100	62.2	23	37.8
Sex						
Male	1255	45.5	862	68.6	257	31.4
Female	1525	54.5	1103	69.9	303	30.1
Family affluence						
Low	920	24.9	732	87.2	60	12.8
Moderate	1285	48.5	901	68.9	285	31.1
High	578	26.6	332	55.2	215	44.8
Family structure						
Both biological parents	1320	52.2	871	62.1	359	37.9
Biological & stepparent	419	19.2	282	67.7	85	32.3
Single parent	516	19.5	383	79.6	74	20.4
Other	269	9.2	211	82.8	24	17.2
Highest education of either parent						
High school or less	979	33.1	748	80.8	105	19.2
Some college	924	39.8	633	66.6	229	33.4
Bachelor or higher	626	27.1	387	58.6	202	41.4
Age at onset of alcohol use						
≥14 years old	1745	76.1	1262	67.1	404	32.9
<14 years old	508	23.9	387	70.9	97	29.1
Urbanicity						
Urban	905	13.9	729	83.0	61	17.0
Non-urban	1629	86.1	1020	66.0	481	34.0

CI: confidence interval; the sum of Ns of DDL and no DDL is not equal to the overall N due to the missing data of DDL variable.

procedures.

3. Results

The sample characteristics overall and by the outcome variable (DDL and no DDL) are shown in Table 1. Of 2525 eligible for licensure during the study period, 887 (38.9%) reported DDL by 1–2 years, 1078 (30.4%) by ≥ 2 years, and 560 (30.7%) reported no DDL (data not shown).

As shown in Table 2 of the eligible NEXT sample (N = 2525), the percentage of participants residing in a state with learner phase restrictions was 20.1% (weighted, hereafter) for ≥16 yrs minimum age, 17.9% for ≥6 month holding period, and 88.6% for ≥30 h required practice. For intermediate phase restrictions, 20.2% of participants had a night driving limit of no later than 10 p.m., and 22.7% and 63.8% had limits on young passengers of ≤2 and ≤ 1, respectively. The percentages of participants in states with different GDL restrictions by the outcome variable (DDL vs. no DDL) are also showed in Table 2.

As shown in Table 3, significant interactions between GDL restrictions at the learner's phase and covariates included race/ethnicity and required practice hours; family affluence and permit holding period; and family structure and permit holding period. Significant interaction between GDL restrictions at the intermediate phase was found between urbanicity and restrictions on young passenger(s). In states that required 30+ hours of supervised practice driving, Latinos (aRRR = 1.55, 95%CI: 1.30, 1.85, p < .001) and Non-Latino Blacks (aRRR = 1.38, 95%CI: 1.10, 1.74, p < .01) were more likely to DDL vs. Non-Latino Whites.

In states that required a permit holding period of <6 month, participants with low (aRRR = 1.61, 95%CI: 1.35, 1.91, p < .001) and moderate (aRRR = 1.45, 95%CI: 1.29, 1.63, p < .001) affluence were more likely to DDL vs. those in high affluence. In states that required a permit holding period of ≥6 month, participants with low affluence (aRRR = 1.33, 95%CI: 1.00, 1.77, p < .05) were more likely to DDL vs. those with high affluence. In states that required a permit holding period of <6 months, participants in single parent households (aRRR = 1.37, 95%CI: 1.00, 1.90, p < .05) were more likely to DDL vs. those living in households with both biological parents.

In states that allowed ≥3 passengers or had no passenger restrictions, participants living in non-urban areas (aRRR = 1.52, 95%CI: 1.06, 2.19, p < .05) were more likely to DDL vs. those in urban areas, and in states that allowed only 1 passenger or no passengers, participants living in non-urban areas (aRRR = 0.67, 95%CI: 0.58, 0.78, p < .001) were less likely to DDL vs. those in urban area.

Supplementary Table 1 shows bivariate association of DDL (DDL vs. no DDL) with demographic variables and GDL restrictions.

4. Discussion

This study examined the associations of DDL among U.S. high school students with state-level GDL restrictions. Our findings are consistent with previous research showing Black and Latino youth are more likely than their White counterparts to delay licensure (Shults and Williams, 2013; Tefft et al., 2014; Williams, 2017) and additionally showed differences by SES.

Our analysis finds that the associations of race/ethnicity and DDL differed by states' requirements for hours of supervised practice driving. Thus, race/ethnicity was differentially associated with DDL depending on the level of GDL restriction, suggesting that more restrictive GDL may be an impediment to licensure among more vulnerable teens.

Several family attributes were found to be significantly associated with GDL factors in our analysis. For instance, in states where the minimum holding period for a learner's permit was <6 months, participants in single parent households were more likely to DDL

Table 2
Prevalence of DDL and GDL restrictions in NEXT participants.

	Overall		DDL		No DDL	
	N	Weighted %	N	Weighted %	N	Weighted %
DDL						
No DDL	560	30.7	–	–	–	–
DDL	1965	69.3	–	–	–	–
Learner's phase minimum permit age						
<16 years	2110	79.9	1664	69.9	446	30.1
≥16 years	415	20.1	301	66.9	114	33.1
Learner's phase permit holding period						
<6 months	1885	82.1	1393	67.2	492	32.8
≥6 months	640	17.9	572	78.7	68	21.3
Learner's phase permit required practice hours						
<30 h	155	11.4	134	79.7	21	20.3
≥30 h	2370	88.6	1831	67.9	539	32.1
Intermediate phase restriction on night driving						
After 10 p.m.	1940	79.8	1436	65.0	504	35.0
10 p.m. or earlier	585	20.2	529	86.4	56	13.6
Intermediate phase restriction on young passenger(s)						
≥ 3 passengers or no restriction	555	13.5	492	83.7	63	16.3
2 Passengers	206	22.7	142	69.7	64	30.3
Zero or 1 passenger	1764	63.8	1331	66.1	433	33.9

Table 3
Association between DDL (delay vs no delay) and demographic variables by GDL restrictions.

	DDL	% DDL	IRR	95% CI		p
Learner's phase: Required practice hours	Race/ethnicity					<.001[a]
< 30 h	Latino vs. Non-Latino White	84.1 vs. 80.5	0.94	0.64	1.37	.73
<30 h	Non-Latino Black vs. Non-Latino White	37.5 vs. 80.5	0.37	0.07	1.94	.22
≥30 h	Latino vs. Non-Latino White	91.6 vs. 55.0	1.55	1.30	1.85	<.001
≥30 h	Non-Latino Black vs. Non-Latino White	84.5 vs. 55.0	1.38	1.10	1.74	<.01
Learner's phase: Permit holding period	Family affluence					<.001[a]
<6 months	Low Affluence vs. High Affluence	84.0 vs. 45.5	1.61	1.35	1.93	<.001
<6months	Moderate Affluence vs. High Affluence	68.4 vs. 45.5	1.45	1.29	1.63	<.001
≥6 months	Low Affluence vs. High Affluence	94.0 vs. 64.0	1.33	1.00	1.77	<.05
≥6 months	Moderate Affluence vs. High Affluence	71.6 vs. 64.0	1.08	0.87	1.33	.50
Learner's phase: Permit holding period	Family structure					<.05[a]
<6 months	Biological/stepparent vs. Both Biological	83.4 vs. 77.6	0.92	0.75	1.13	.41
<6 months	Single Parent Vs. Both Biological	85.5 vs. 77.6	1.37	1.00	1.90	.05
≥6 months	Biological/stepparent vs. Both Biological	66.2 vs. 59.8	1.11	0.95	1.28	.17
≥6 months	Single Parent Vs. Both Biological	79.4 vs. 59.8	1.17	0.74	1.84	.48
Intermediate phase restriction on young passenger(s)	Urbanicity					<.001[a]
≥ 3 passengers or no restriction	Non-urban vs. Urban	97.0 vs. 66.9	1.52	1.06	2.19	<.05
2 Passengers	Non-urban vs. Urban	69.0 vs. 87.7	0.85	0.64	1.14	.26
Zero or 1 passenger	Non-urban vs. Urban	60.3 vs. 97.6	0.67	0.58	0.78	<.001

IRR: incidence rate ratios. Interactions between demographic variables and restrictions were examined. For significant interactions, the association between DDL and demographic variables by level of GDL restriction are reported in this table.
[a] Indicate the overall p-value for the interaction.

compared to participants living in a household with both biological parents. The learner's stage is designed to increase driving practice in as safe of an environment as possible. This includes supervised driving often completed by parents (Williams et al., 2012b). In a single parent compared to a two-parent household, potentially more limited time and vehicle availability/access could make it considerably more difficult to provide dedicated supervised practice driving. Further, it is conceivable that a single parent may find it more challenging to confidently assess a teens' "developmental" readiness for driving at a younger age (i.e., ≤16 years) as compared to two parents making a collective-shared decision. Some parents may view obtaining a permit at these earlier ages to be unnecessary, inconvenient, fraught with parenting challenges and/or economic strains, and, ultimately, a developmental milestone that can wait (Simons-Morton and Hartos, 2003).

Another family attribute which was significantly associated with GDL factors in our analysis was family affluence. As suggested in previous studies, economic circumstances are inextricably tied to licensure such that family economic burdens may result in the delay of licensure (Schoettle and Sivak, 2014; Shults and Williams, 2013). We found that participants with low family affluence were more likely to DDL compared to those with high family affluence no matter the length of the holding period. This could possibly be due to the financial demands of vehicle ownership (Tefft et al., 2013). Moreover, previous research in the U.S. covering 1996–2010 showed that the U.S. economic recession (December 2007–June 2009) led to reductions of high school senior licensure rates (Shults and Williams, 2013). In this study, low-income status showed similar effects on DDL (i.e., greater likelihood of DDL). Therefore, the economic cost of licensure may be a dominant factor for youth and their families with low affluence as they determine the timing of licensure.

The geographical attribute was differentially associated with DDL depending on the level of GDL restriction at the intermediate phase. In states where GDL was less restrictive (i.e., allowing 3 young passengers or no restriction on young passengers), non-urban teens were more likely to DDL compared to urban teens. In contrast, when comparing non-urban to urban teens living in states with more restrictive GDL (i.e., allowing only 1 passenger or no passengers), non-urban teens were less likely DDL. We speculate, that in general, non-urban families could have a higher financial burden to bear compared to urban families when it comes to teen licensure. As a result, the cost of licensure and associated vehicle costs (e.g., gas, registration, insurance) may heighten the burden of family finances among non-urban teens, resulting in DDL. In the states that have lower passenger restrictions, teens may choose to ride with peers which may reduce the immediate necessity for their own licensure. Coupling the financial burden of licensure along with an attenuated necessity of licensure, given the availability of carpooling with a driving peer, could facilitate an increase in DDL among non-urban adolescents. In contrast, despite the financial strain, some non-urban adolescents in states that allowed only 1 or no passenger may not be able to ride with peers due to strict passenger restrictions. Consequently, these teens may be forced to pursue timely licensure, avoiding DDL, so they can meet important transportation needs (e.g., school, sport/social activities, employment). More research is needed to explore this area and to bring greater clarity to these scenarios.

Graduated Driver Licensing (GDL) programs in different countries have been effective in reducing crashes and injuries among young drivers, where licensure is obtained at older ages (Begg and Stephenson, 2003; Christie et al., 2017; Senserrick and Williams, 2015; Williams, 2017). Given that GDL in the U.S. has mainly applied to those under 18 years of age, it is unclear if GDL could influence crash risk if more widely applied to older novice drivers. A recent study conducted in the state of Queensland, Australia, evaluated the effect of a strengthened GDL program that applied some GDL requirements to those under 25 years found substantial declines in all novice driver crashes and serious crashes as well as the resulting deaths and/or injuries (Senserrick et al., 2018). These findings argue that requiring GDL of those at a later licensing age improve novice drivers' performance, increase experience, and reduce fatal crashes. However, despite these implications, in the U.S. we know little of how an extended-age GDL program might affect differences and disparities among not only DDL but also definitively among unlicensed groups of older novice drivers.

Finally, while not evaluated in this study, an additional area that we believe is worthy of highlighting in the context of DDL is state level requirements for driver education. For some teens and their families, given their contextual setting (i.e., state residence, household income), driver education requirements may influence decisions related to DDL. Taking into consideration that enrolling and participating in a driver education course takes time and money, it's probable that such a requirement could have a disproportionate impact on lower SES teens (e.g., consideration of interaction effects between affluence, race/ethnicity, family structure). A recent U.S. survey of young adults ages 18–24 years provided some suggestive evidence of the impact of driver education requirements on DDL. In this U.S. representative sample study, there were similar licensure rates for those youth that became licensed at ≤16 years regardless of state-specific driver education requirements. However, licensure rates at age 17 years were found to be higher in states that did not require completion of a driver education course before the new driver could be licensed. Further, in states that required a driver education course as prerequisite for licensure for youth age <18 years, but not for those new drivers ≥18 years, licensure at ≥ 18 years was more common. Interestingly, the study found that the largest proportion of "not-yet-licensed" new drivers were those that lived in states that required the completion of a driver education course for drivers ≥18 years. (Tefft and Foss, 2019). While we firmly believe that driver education is of major importance for novice drivers of any age, we also believe that additional study of the effects of state-level driver education course requirements on DDL would be beneficial in understanding identified disparities in delayed licensure.

We recognize that our study has limitations. First, the school-based recruitment limits the generalization to youth in school at 10^{th} grade but not those who were not enrolled in school. Second, it would have been ideal to know the exact date on which participants obtained their driver's license so that DDL could be more accurately estimated. However, these data were available in the NEXT survey data. As a result, we were limited to calculating and approximating the DDL variable. Third, only a limited number of covariates were collected and analyzed. This may exclude other important factors influencing DDL (e.g., youth time conflict, parents' schedule, youth needs for transportation). Fourth, although the influence of economic factors is acknowledged, the information on the overall cost of licensing in different states is missing. Therefore, it is unclear whether and/or how a more accurate state-based economic burden measure (e.g., state-level licensing costs) may influence DDL.

5. Conclusion

Differences and disparities in DDL among racial/ethnic groups exist. This could be limiting the beneficial GDL safety effects of teen novice drivers with potentially more negative results among more vulnerable youth groups. Our findings cohere with previous research attributing some part of DDL to the economics of licensure. Furthermore, important family level factors were related to DDL. Additional investigation is warranted to disentangle the complexity of factors contributing to DDL. Advancing our understanding of disparities in DDL could inform policy and enhance efforts to further prevent injury among more vulnerable teen groups.

CRediT authorship contribution statement

Federico E. Vaca: Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Visualization, Funding acquisition. **Kaigang Li:** Investigation, Methodology, Formal analysis, Writing – original draft, Writing – review & editing, Visualization. **James C. Fell:** Writing – review & editing. **Denise L. Haynie:** Investigation, Resources, Writing – review & editing, Funding acquisition. **Bruce Simons-Morton:** Investigation, Resources, Writing – review & editing, Funding acquisition. **Eduardo Romano:** Conceptualization, Writing – review & editing, Funding acquisition.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jth.2021.101068.

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